XRPL Validator — Daily Engineering Report · 2026-07-20

Period: 2026-07-20 00:00 to 24:00 US Eastern (2026-07-20 04:00 to 2026-07-21 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04); data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-07-20): 1.00000 (0 missed of 21,569). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(xrpl-ledger): issuer TickSize in offer placement — corpus 95.0%** (`a5db381`, xrpl-validator) OfferCreate ignored the issuer-published TickSize. rippled rounds the requested rate UP to N significant digits (Quality::round) and re-derives the non-exact side — TakerPays for tfSell, TakerGets otherwise — BEFORE crossing, so both the crossing and the placed remainder use the adjusted amounts. Mainnet tx 9870DA80 asked to sell 8539920 drops for STX (issuer TickSize 6) and rippled stored 8539914 with no …
- **fix(xrpl-ledger): route strand output to the destination — corpus 95.6%** (`bb1eefa`, xrpl-validator) A cross-currency Payment credited the acquired asset to the SENDER and then forwarded it to the destination. rippled delivers the strand's output straight to the destination: routing through the sender materializes an intermediate trust line it never creates — and when the destination IS the issuer (mainnet tx AAA6EB389D3A: RIBBLE paid to rG7jT6D4, its own issuer), the IOU is redeemed at the maker/AMM line rather …
- **fix(xrpl-ledger): trust-line lifecycle — receiver-only reserve, NoRipple, zero-balance deletion — corpus 95.6%** (`d547205`, xrpl-validator) Three faithful-port fixes in line_adjust, from rippled rippleCreditIOU / trustCreate: - trustCreate charges the reserve to the RECEIVER only. We bumped BOTH parties' OwnerCount, so the issuer accrued phantom reserves. The line still joins both owner directories; only one OwnerCount moves. - New lines get NoRipple on the receiver's side unless their account has DefaultRipple. This flag is also part of the deletion …
- **feat(xrpl-ledger): tfSell crossing semantics — sell the whole TakerGets — corpus 96.3%** (`3a3f337`, xrpl-validator) A tfSell offer sells its ENTIRE TakerGets even if that acquires more than TakerPays; TakerPays is only a minimum. Our crossing engine capped each fill at rem_pays (the wanted amount) and terminated the walk once

it was reached, so a tfSell that could fully sell only by taking the surplus was left with an unsold remainder — and a FillOrKill/ImmediateOrCancel sell was then wrongly killed. Mainnet tx 2042F8CE / …

- **fix(xrpl-ledger): sell-side offer funding check — tecUNFUNDED_OFFER — corpus 96.8%** (`870eefd`, xrpl-validator) An OfferCreate can only sell what the account actually holds of the TakerGets asset. Our crossing engine checked the MAKER funding but never the taker: after the tfSell fix started crossing the full TakerGets, five arb offers from one account selling IOUs it holds NONE of (CLARITY, XALPHA) crossed and succeeded, where mainnet returns tecUNFUNDED_OFFER fee-only. Verified: the account CLARITY line was absent at seq-1.
- **fix(xrpl-ledger): faithful OfferCreate preclaim funding — tecUNFUNDED_OFFER — corpus 96.8%** (`5b1d386`, xrpl-validator) OfferCreate::preclaim returned the generic TxResult::Unfunded (tecUNFUNDED) and rejected whenever balance < the FULL sell amount. Both are wrong: rippled unfunds only when accountFunds(TakerGets) <= 0 (for XRP: balance minus reserve — partially funded offers still cross) and the code is tecUNFUNDED_OFFER. The old check was also XRP-only.

Independence program

No independence record for UTC day 2026-07-20; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **On-Ledger Positive Validator Reputation Score** (consensus) Add a singleton `ValidatorReputation` ledger object (keylet analogous to NegativeUNL's, reserve-free, capped to current UNL size) populated once per f
- **Time-Limited NFT Lease Via Session Keys** (nft) Add a new `NFTokenLease` transaction that creates an owner-reserved `ltNFTOKEN_LEASE` object (keylet derived from NFTokenID + sequence) binding an exi
- **Revocable Hash-Commitment Attestation Registry** (audit) Add a new `ltATTESTATION` ledger object (owner-reserved by the issuer, `keylet::attestation(issuer, sequence)`) created via `AttestationSet`, storing

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
07-20 14:24	—	xrpl-validator	`a5db381`	fix(xrpl-ledger): issuer TickSize in offer placement — corpus 95.0%
07-20 16:04	—	xrpl-validator	`bb1eefa`	fix(xrpl-ledger): route strand output to the destination — corpus 95.6%
07-20 20:28	—	xrpl-validator	`d547205`	fix(xrpl-ledger): trust-line lifecycle — receiver-only reserve, NoRipple, zero-balance deletion — corpus 95.6%
07-20 22:19	—	xrpl-validator	`3a3f337`	feat(xrpl-ledger): tfSell crossing semantics — sell the whole TakerGets — corpus 96.3%
07-20 23:49	—	xrpl-validator	`870eefd`	fix(xrpl-ledger): sell-side offer funding check — tecUNFUNDED_OFFER — corpus 96.8%
07-21 01:19	—	xrpl-validator	`5b1d386`	fix(xrpl-ledger): faithful OfferCreate preclaim funding — tecUNFUNDED_OFFER — corpus 96.8%

Times in UTC. A commit appears on the day it was committed to main or merged into it.